Phase II - Experiment Plan (pre-specified protocol)

Project: Comparative Analysis of AI-Assisted and Manual Data Labeling Accuracy Across Scientific Datasets **Phase:** II (Experiment plan) | **Written:** 2026-10-28, **before** any accuracy was computed | **Due:** 30 Oct 2026 **Inputs:** `gap_matrix.md`, `research_decisions.md`, `literature_to_experiment_bridge.md` (D-008 to D-012), `feasibility_novelty_matrix.md` (D-013 to D-017)

Tags: VERIFIED = read in a file or page opened in this project. INFERRED = my reasoning, stated. UNKNOWN = not checked; the check is named. PROPOSED = a design choice that needs your confirmation. **Standing rules (D-003):** no invented papers, links, runs or metrics.

> **What I looked at before writing this plan.** Only the file structure of CIFAR-10H: the file names, the 13 column names, the row count, and that `cifar10h-counts.npy` has 10,000 rows with 47 to 63 judgments each. I computed no accuracy and no agreement. Every rule below that could favour an outcome (reference label, primary contrast, budget) is fixed here first.

1. Verified inputs that shape the plan

Item	Finding	Tag
CIFAR-10H licence	**Resolved:** `LICENSE.txt` in the repository states Creative Commons BY-NC-SA 4.0 (non-commercial; credit; indicate changes; share-alike). Cloned commit 389f22d (23 Mar 2026). Closes V7 and D-016.	VERIFIED
CIFAR-10H files	`cifar10h-counts.npy` and `cifar10h-probs.npy` (10,000 x 10), `cifar10h-raw.zip` -> `cifar10h-raw.csv`: 539,910 rows x 13 columns (annotator_id, trial_index, is_attn_check, true_category, chosen_category, true_label, chosen_label, correct_guess, cifar10_test_test_idx, image_filename, subcategory, reaction_time, time_elapsed); 2,571 annotator IDs; no missing values.	VERIFIED
Row count vs documented count	The dataset page states 511,400 classifications; the raw file has 539,910 rows, so it must include attention-check trials and possibly excluded workers. The code must reproduce `counts.npy` from the raw file as a validation check (Section 8).	VERIFIED (numbers); explanation INFERRED
CIFAR-10 images	**Not reachable from this workspace** (host returned a proxy refusal). Needed for the model arm.	VERIFIED (refusal)
Deep-learning packages	**torch and torchvision cannot be installed here** (pip found no matching distribution). scikit-learn 1.9.1, pandas, scipy, numpy are present. Hardware here: 2 CPU cores, 7 GB RAM, no GPU.	VERIFIED
Consequence	The label-only parts (manual arm, noise floor, reference-rule comparison) and the routing code can be run and validated here. The **model arm on real CIFAR-10 images must be run on your machine or a notebook**.	INFERRED

2. Research question and hypotheses

Question (from D-013, tightened): On the 10,000 CIFAR-10 test images, when a classifier pre-labels every image and a human then corrects a fixed fraction b of them, how much does it matter **which** images are sent to the human, and does the answer change when the human is a single imperfect annotator instead of a perfect reviewer?

Why the reviewer type matters (INFERRED). With a perfect reviewer, the gain from routing is just how many of the model's errors land in the reviewed set. Any reasonable uncertainty score will beat random there, so a positive result would confirm what is expected and not be a finding. With a single annotator, the images the model finds hardest are also the ones where people disagree (CIFAR-10H's own purpose), so the benefit of routing may shrink or even reverse. That is the open part.

ID	Hypothesis	Role	What would count against it
H1	Perfect reviewer: entropy-ranked review gives higher final accuracy than random review at every budget b.	Pipeline check, upper bound. Not claimed as a finding.	Any budget where the paired CI includes zero or is negative: treat as a bug until traced.
H2 (primary)	Single-annotator reviewer: entropy-ranked review at b = 0.20 gives higher final accuracy than random review at b = 0.20.	Primary test	Paired 95% CI of the difference includes zero or is below zero. A null or negative result is a reportable outcome.
H3	Single-annotator review of the most uncertain items can **lower** accuracy below model-only at some budget (humans worse than the model on the hardest items).	Secondary, exploratory	Not observed at any budget.
H4	The accuracy difference between the classifier and a single random annotator is larger than the annotator-to-annotator spread.	Secondary (G5 noise floor)	Difference lies inside the spread.

H2 is the only confirmatory test. All others are descriptive and reported with intervals, no p-value hunting.

3. Definitions (fixed)

- **Items:** the 10,000 CIFAR-10 test images, indexed by `cifar10_test_test_idx`.
- **Reference label r_i (primary):** the original CIFAR-10 test label (`true_label` in the raw file). *Reason:* it is independent of the annotators being evaluated. **Sensitivity analysis:** the human plurality label (argmax of `counts`; ties broken by the lower class

index, reported as a count of ties). Both are fixed now. Which is better is contested (the "Pervasive Label Errors" paper is only known to me by title, so I make no claim about CIFAR-10 test errors).

- **Pre-label:** the classifier's argmax class, with probability vector p_i.
- **Budget b:** fraction of items sent to review, k = round(b x 10,000). Grid: 0.05, 0.10, 0.20, 0.40. **Primary budget: 0.20.**
- **Reviewer types:**
- *Perfect:* returns r_i.
- *Single annotator (primary):* returns one label drawn uniformly at random from the item's raw judgments (attention-check trials removed).
- **Final label:** the reviewer's label for reviewed items, the pre-label for the rest.
- **Outcome:** accuracy of final labels against r_i.

Arms

Arm	Selection rule for the k reviewed items	Notes
none	nobody reviewed	Model-only baseline
random	k items uniformly at random	The control
entropy	k items with highest predictive entropy	**Primary uncertainty score**
margin	k items with smallest gap between top two probabilities	Secondary
least-confidence	k items with lowest top probability	Secondary
oracle	the model's actual errors first, then random	Ceiling for the perfect reviewer; not achievable
manual	no model; one annotator's label per item	G1 arm

Change from Stage 2 (D-018, PROPOSED): "fixed-fraction review" is dropped. In Stage 2 it was never defined apart from random, and any fixed-fraction rule that ignores the model has the same expected accuracy as random review. Keeping it would add an arm that cannot differ. If you meant a different rule (for example a confidence threshold), say so and I will add it.

Change from Stage 2 (D-019, PROPOSED): the single-annotator reviewer moves from optional to primary, for the reason in Section 2.

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4. Model arm

Element	Choice	Tag
Features	Frozen pretrained image backbone, one forward pass over 50,000 train and 10,000 test images, cached	PROPOSED
Candidate backbone	A small ImageNet-pretrained ResNet from torchvision, images resized to the backbone's input size	PROPOSED; **weights licence and download are UNKNOWN** (not opened; check before use)
Classifier	Multinomial logistic regression on the cached features, trained on the CIFAR-10 training set only	PROPOSED
Calibration	None in the primary analysis. Report expected calibration error as a descriptive number.	PROPOSED
Leakage rule	The test set is never used for training, model selection, or choosing hyperparameters. Hyperparameters are chosen on a held-out 10,000 split of the training set.	PROPOSED
Seeds	10 seeds. Each seed: classifier trained on a random 80% of the training set; random-arm draws; single-annotator draws.	PROPOSED
Backbone fallback	If the torchvision weights cannot be used, train a small CNN from scratch on CIFAR-10 (more compute, lower reproducibility) or use another openly licensed backbone after reading its licence	INFERRED

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5. Statistics

- **Primary estimate:** mean difference in final accuracy, entropy minus random, single-annotator reviewer, b = 0.20, averaged over seeds.
- **Uncertainty:** paired bootstrap over test items (2,000 resamples), with arms evaluated on the same resampled items; 95% percentile interval. Seeds are reported separately as a spread (min, median, max), because item resampling and seed variation are different sources of noise.
- **No multiplicity correction** on descriptive results; the single confirmatory test is H2.
- **Noise floor (H4):** per-annotator accuracy against r_i, over the items each annotator saw (about 200 each per the dataset page), reported as mean, SD and 5th/95th percentiles; and the spread of single-annotator accuracy across 1,000 random one-label-per-item draws.
- **Stop rule:** all arms and budgets are reported, including losses. No arm is dropped after seeing results. Any change to this plan after the first model-arm run is logged as a deviation with the reason.

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6. What this experiment can and cannot support

Can support	Cannot support
How routing rules compare at equal budget on one natural-image set, with a perfect	Any claim about **scientific** datasets (title) - needs a second

and with a single-annotator reviewer	domain (V10)
Whether a model beats a single crowd annotator on these items, relative to annotator spread	Time savings, fatigue, or anchoring: the reviewer is simulated and never sees the model's label
A null result is informative	That uncertainty routing works for text or segmentation

Threats to validity (INFERRED): 1. The simulated reviewer does not see the pre-label, so it cannot be anchored by it (G2 is untouched). Results are optimistic for AI-assisted review. 2. CIFAR-10H annotators were crowd workers on 32x32 images, not domain experts. 3. A reviewer's drawn label is independent of earlier draws; a real person has habits. 4. The reference label may itself be wrong for some images; the sensitivity analysis shows how much it matters. 5. One classifier family. A different backbone could change the ranking quality. 6. Feature extraction and the 80% training subsample capture only part of classifier variance.

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7. Resources, time and what must run where

Task	Where	Time (INFERRED)
Label-only analysis (manual arm, noise floor, reference sensitivity), routing logic tests, pipeline check on a small bundled dataset	**This workspace (done in Phase III)**	Minutes
CIFAR-10 download, feature extraction, model arm, 10 seeds, bootstrap	**Your machine or a notebook**	Feature extraction dominates; **unmeasured**. Time one 500-image batch first.
Write-up	You	1 to 2 days

Deadline reality (INFERRED): Phase II and III are due 30 Oct. The model arm needs one notebook session; if you cannot run it by then, hand in the plan, the validated code, the label-only baseline and an explicit statement that the model arm is pending.

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8. Phase III validation gates (pass before any model-arm number is trusted)

Gate	Check	Pass criterion
G-1	Routing functions on hand-made probabilities with known answers	All unit tests pass
G-2	Perfect-reviewer accuracy with `oracle` arm	Equals the closed-form value min(1, base acc + k/N)
G-3	Perfect reviewer, b = 1.0	Accuracy = 1.0 for every arm
G-4	Raw file -> counts	Reconstructed counts equal `cifar10h-counts.npy` exactly after attention-check and exclusion rules; if not, stop and trace
G-5	Bootstrap	On a constructed pair with a known difference the interval covers it; with identical arms it covers zero
G-6	Determinism	Same seed gives identical output
G-7	End-to-end pipeline on a real classifier with a bundled dataset (no download)	Runs, results consistent with G-2/G-3, clearly labelled as **not a Study B result**

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9. Decision log additions

ID	Decision	Status	Next action
D-018	Drop the undefined "fixed-fraction" arm	PROPOSED	You confirm or define it
D-019	Single-annotator reviewer becomes the primary reviewer	PROPOSED	You confirm
D-020	Reference label = original CIFAR-10 label; human plurality as sensitivity	PROPOSED	You confirm **before** the model arm runs (closes V8)
D-021	Primary contrast: entropy vs random, single-annotator reviewer, b = 0.20; H2 only confirmatory test	PROPOSED	You confirm
D-022	CIFAR-10H licence CC BY-NC-SA 4.0: results and code that adapt it must say what changed and carry the same licence; non-commercial use only	VERIFIED (licence text)	Check your course or venue allows this
D-023	Model arm to run outside this workspace; label-only parts and code validated here	PROPOSED	You run it; send me the output files

Still open from earlier stages: backbone and its licence (V-new), the scientific second dataset (V10, V11), hardware and exact deadline (O-5, O-6), authorship of the source review (D-004).

Bridge: original papers -> selected method -> executable baseline -> planned test -> novelty and feasibility gate -> pre-specified protocol (this file) -> validated code (Phase III).